Chandrayaan-2, errors were programmed to get sorted as and when they arose and not wait for the coasting phase.

Meanwhile, at the SAC in Ahmedabad, Desai had all the 11 sensors that monitor the major parameters of Vikram and the eight cameras that act as its eyes tested rigorously and ruggedised. This included vast improvements in the Lander Hazard Detection and Avoidance Camera (LHDAC) that kicks into action in the final minutes of Vikram's descent. The craft hovers over the landing spot at a height of 150 metres and waits till the pictures from the LHDAC confirm that there are no boulders, craters or other obstacles before descending further. Redundancies were also built into the cameras to ensure that if one fails, the other would assume its place. They even simulated the lunar surface by transporting sand from Salem similar to lunar regolith and created a mini landing strip complete with craters and boulders to test the cameras.

There were other innovative solutions. With the Chandrayaan-2 lunar orbiter a success, the team dispensed with the need to have a long-duration orbiter for Chandrayaan-3. So its orbiter was made lighter and the lander heavier by over 250 kg. This allowed them to fix more back-up instruments to build redundancy in the event that any failed. Much of this capacity was used to carry extra fuel to meet any exigencies and deviations on the planned descent path. On the previous occasion, there was just a "touch-and-go" amount of fuel, which cramped the autonomous decision-making process. Now, they could carry an extra 182 kg of fuel. With the Vikram lander getting heavier, the fifth engine in the Chandrayaan-2 lander for use in its final descent phase was also dispensed with. The current lander, therefore, has four engines, two of which kept firing till touchdown. The removal of the fifth engine from the lander's belly made its centre of gravity more stable while also raising the ground clearance, which was essential if the craft encountered undulating terrain. Vikram's leg pads, too, were strengthened to absorb twice the force if needed. These leg pads are made of aluminium in a unique honeycomb pattern that gets crushed on impact, absorbing most of the shock of the fall.

WITH AN ORBITER ALREADY IN PLACE, IT WAS DISPENSED WITH, MAKING THE LANDER LIGHTER. A FIFTH ENGINE TOO WAS REMOVED

Another issue was related to the area that had been identified for Chandrayaan-2's landing. It was a narrow band of 500 metres by 500 metres, which made manoeuvring even more complex. To circumvent this problem, the Chandrayaan team used its existing orbiter to map the landing region thoroughly and expanded the landing zone to 4 km by 2.4 km. As a team member put it, "The comparative difference is like landing on a football field instead of a tennis court."

All the niggles thus sorted and every head of systems involved in Chandrayaan saying they were satisfied that they had made failsafe improvements, Somanath gave the clearance to launch the spacecraft on July 14, 2023. After the LVM3 launched the Chandrayaan-3 propulsion module and lander perfectly into orbit, Mohana Kumar, the mission director, said, "This successful launch is the result of the penance of many ISRO centres across the country." Indeed it was.

Veeramuthuvel, who was at the launch pad, knew it was now over to him and his team to prove its success. He had worked hard for it and become so passionately involved that URSC's Sankaran would joke, "We were worried that he would strap himself to LVM3 and personally escort Vikram to the moon." It would take 40 days before the team would attempt to land Vikram after it successfully detached from its module. On D-Day, both Veeramuthuvel and Kalpana were huddled over their consoles. "The first thing we watched for," says Kalpana, "was the rough brake phase, which went beautifully. Our hearts beat faster as we approached the phase where it had failed the last time." Once Vikram crossed that rubicon, Veeramuthuvel would look at Kalpana and smile. "Once that went through," he says, "we were confident that it would land perfectly. Thousands of people had worked on the mission and had done everything that was humanly possible to ensure there was no failure."

Not far from them sat Somanath with other top ISRO team members keeping a close watch on the console. When Somanath saw the five green squares flash on his console, which indicated that Vikram had accomplished a soft landing on the moon and all systems were functioning fine, he would shake hands with his colleagues, walk across to Veeramuthuvel and hug him. After that, he went on the dais to inform the prime minister, who was watching the event live on a television screen in Johannesburg, where he had gone for a BRICS meeting. As Modi happily waved the Indian flag, Somanath told him, "Honourable Prime Minister, we have achieved soft landing on the moon. India is on the moon." It was, if one were to rephrase what Neil Armstrong said in 1969, another small step for humanity, but a giant leap for India. ■